

# Mohamed Hamada Ghanem

Mobile Software Engineer — Flutter & Dart Specialist — Cross-Platform Development

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## Professional Summary

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Results-driven **Mobile Software Engineer** specializing in **Flutter and Dart**, with hands-on experience architecting high-performance, cross-platform applications for Android and iOS. Expert in implementing **Clean Architecture**, advanced state management (**BLoC, Freezed**), and secure **RESTful API** integrations. Proven track record of engineering scalable solutions, including comprehensive healthcare platforms, multi-role access systems (RBAC), and integrating AI-driven backend services. Adept at translating complex UI/UX designs into responsive, production-ready, and maintainable applications.

## Core Competencies & Technologies

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**Mobile Technologies:** Flutter SDK, Dart, Android, iOS, Material Design, Cupertino

**Architecture & State Management:** Clean Architecture, MVVM, MVC, BLoC (Cubit), Provider, Freezed, GetIt (DI)

**Backend & Data Integration:** RESTful APIs (Dio/Http), JSON Parsing, Firebase (Auth, Firestore, FCM), JsonSerializable

**Local Storage & Databases:** SQLite, Hive, SharedPreferences

**Tools & Methodologies:** Git, GitHub, Postman, Figma to Code, Agile Methodologies, VS Code, Android Studio

**Specialized Domains:** Role-Based Access Control (RBAC), AI/ML Integration, Localization (i18n), Dynamic Theming

## Professional Experience

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### Mobile Software Engineer (Flutter) – Project Based Experience

2024 – Present

- **Architected and deployed** high-performance cross-platform mobile applications using the **Flutter SDK** and **Dart**.
- **Engineered scalable app architectures** utilizing **Clean Architecture** and **Dependency Injection**, resulting in highly maintainable and testable codebases.
- **Implemented advanced state management** solutions using **BLoC** and **Cubit**, combined with **Freezed** for robust, immutable data modeling.
- **Integrated complex RESTful APIs and AI-driven services**, enabling real-time medical image processing and seamless cloud data synchronization.
- **Designed Role-Based Access Control (RBAC) systems**, securely managing multi-tier user permissions across various applications (e.g., Admin, Supervisor, Doctor, Patient).
- **Optimized UI/UX performance** by minimizing widget rebuilds, implementing lazy loading, and ensuring fluid 60-FPS rendering across diverse screen sizes.

## Key Engineering Projects

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### Educational & Religious Courses Platform (Soon on Google Play)

Flutter — BLoC — Freezed — Clean Architecture — REST API — GetIt

- **Engineered** a comprehensive multi-role e-learning platform (Admin, Supervisor, Student) with secure access controls and dynamic, role-specific dashboards.
- **Developed** an advanced Admin portal for comprehensive content management, supporting multimedia lessons (Video/Audio/PDF), live sessions integration, and dynamic quizzes.
- **Architected** a robust assignment grading and feedback workflow, empowering supervisors to monitor and evaluate student progress efficiently.
- **Delivered** a premium user experience utilizing **ScreenUtil** for responsiveness, dynamic Light/Dark theming, and full Arabic/English **Localization**.

### AI-Powered MRI Contrast Enhancement Application

Flutter — Medical Image Processing — AI Integration — REST API

- **Developing** an innovative healthcare solution that leverages AI to transform standard **MRI scans** into Contrast-Enhanced (Dye-based) medical images.
- **Integrating** Python-based Deep Learning REST APIs to process and render high-resolution medical imagery directly within the mobile client.
- **Designing** an intuitive, accessibility-focused interface tailored for medical professionals to securely upload, view, and analyze patient scans in real-time.

### **Comprehensive Healthcare & Medical Booking System**

*Flutter — Role-Based Access Control — BLoC — UI/UX*

- **Architected** a robust medical scheduling application supporting three specialized user roles: Doctor, Patient, and Nurse.
- **Built** secure authentication workflows and personalized dashboards for streamlined appointment management and medical history tracking.
- **Utilized BLoC** to ensure robust handling of complex booking business logic and prevent state concurrency issues.

### **HD Wallpapers App (Pexels API Integration)**

*Flutter — REST API — Media Caching*

- **Developed** a dynamic media application integrated with the **Pexels API** for real-time wallpaper fetching.
- **Implemented** efficient image caching mechanisms and pagination strategies to optimize network usage and enhance scroll performance.

### **Premium Hotel Booking Application**

*Flutter — Custom Animations — API Integration*

- **Created** a responsive hotel reservation platform featuring complex search algorithms, filtering, and booking workflows.
- **Focused** on a high-end UI/UX design, incorporating custom micro-animations to enhance the user journey from browsing to checkout.

## **Education**

**Bachelor of Information Technology**

*Expected 2026*

Delta Technology University – Egypt

## **Certifications & Training**

- The Complete Flutter Development Bootcamp with Dart
- Advanced Flutter: BLoC, Freezed, and Clean Architecture
- Mobile App Development with Flutter & Firebase

## **Soft Skills**

- Advanced Problem Solving & Technical Debugging
- Cross-functional Team Collaboration & Communication
- Agile Development & Rapid Adaptability
- Strong Attention to Detail & Clean Code Principles

## **Languages**

Arabic (Native) — English (Professional Working Proficiency)